

Total No. of Questions : 10]

SEAT No. :

P3054

[Total No. of Pages : 4

[5354]-543

B.E. (Mechanical) (End Semester)
MECHANICAL SYSTEM DESIGN
(2012 Pattern)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) Give significance of 'Node Method' of optimization to explain why structure diagram with minimum nodal sum is optimum. **[4]**
- b) Tensile strengths of population of 1000 tie rods are normally distributed with mean of 500MPa and std. deviation of 50MPa. Find number of connecting rods with strengths less than 450MPa. Use table 1 for values of area from $z = 0$ to z . TABLE 1 **[6]**

OR

- Q2)** a) Prove that the difference between number of teeth of successive gears in change gearbox must be more than four. **[4]**
- b) Give formulae for calculating Mean and Standard deviation of resultant population if two individual populations are to be added, multiplied and divided. **[6]**
- Q3)** a) What is the need of tension take up device in a belt conveyor? Name any four take up devices. **[4]**

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- b) Find required width of a flat belt conveyor if it is designed to transfer powdered material with density 2 tons/m³ at rate of 1.33 tons/hr, with speed of 1.75m/s. (Assume surcharge factor as 2.35×10^{-4}) [6]

OR

- Q4)** a) Explain advantages and limitations of troughed conveyors over flat belt conveyors. [4]

- b) Mr. X is required to transmit the coal from coal storage to boiler house by conveyor so suggest him suitable type of conveyor and explain different resistive forces for estimating power requirement of conveyor. [6]

- Q5)** a) Seam less cylinder with storage capacity 0.025m³ is made of 40C8 with permissible strength of 500MPa. It is to be used for storing liquid at 15MPa Pressure. Its length is twice of ID. If factor of safety required is 2, determine dimensions of cylinder. [9]

- b) Derive the expressions to find principal stresses at the inner surface of a thick cylinder. [9]

OR

- Q6)** a) A pressure vessel of 2m ID and 12mm shell thickness is designed for 0.825MPa pressure. It has nozzle of 300mm ID, 12mm thickness and 200mm length, out of which 15mm is inside the shell. S_y is 225 MPa. Corrosion allowance is 4mm and weld efficiency is 0.85 Assuming FOS as 1.5, decide dimensions of reinforcing pad. [9]

- b) What is the need of Autofrettage? Explain any one method of autofrettage in detail along with the supporting sketch. [9]

- Q7)** a) Four stroke diesel engine has following specifications: B Power = 5 kW, Speed = 1200 RPM, IMEP = 0.35 N/mm², efficiency = 80%, Gas pressure 3.15MPa, Permissible stress for cylinder = 42 N/mm². Determine cylinder dimensions & cylinder head thickness. [8]

- b) Explain the procedure of designing connecting rod. [8]

OR

Q8) a) Four stroke diesel engine has following specifications: Cylinder bore=85mm, Gas pressure 3MPa, Allowable bearing pressure for skirt=0.4MPa, ratio of side thrust to gas load=0.1, Width of top land=20mm, width of grooves=2.75mm, piston rings=4, thickness of rings=3mm. Calculate length of skirt and piston length. [8]

b) Explain the procedure of designing crank shaft and crank pin. [8]

Q9) a) A tensile bar of length 400mm is subjected to constant tensile force of 4000N. If the factor of safety is 2, design the bar diameter, using Johnson's method, with the objective of minimizing material cost by using optimum material from the list given in Table 2. What will be the cost of the component? [8]

Material	Density (ρ) Kg/m ³	Cost (c) Rs/Kg	Syt N/mm ²
Steel	7800	28	400
Aluminum Alloy	2800	140	150
Titanium Alloy	4500	2200	800

Table : 02

b) What are the principals of design of castings and forgings? [8]

OR

Q10) a) How to identify whether an optimum design problem is based on normal specifications or redundant specifications? What is the difference in the design procedure of problems based on these two specifications? [8]

b) Explain the concepts of design for assembly and design for safety. [8]

z	0	1	2	3	4	5	6	7	8	9
0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998
3.5	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998	0.4998
3.6	0.4998	0.4998	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.7	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.8	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999	0.4999
3.9	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000

Table 01: Areas below Normal Distribution Curve for 0 to Z.

